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Candidate Rotation Periods for 34 Main-belt Asteroids from Gaia DR3 Sparse Photometry

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ABSTRACT

I searched Gaia DR3 sparse photometry for asteroid rotation periods with a periodogram that combines season offsets, a linear phase term and a second-order Fourier series. Among the numbered asteroids that Gaia observed most often, I analysed the 300 highest-ranked with a known period (validation set) and the 298 highest-ranked without one (target set). On 337 confident results that can be compared with published periods, 80% agree within 3%. Accuracy is 81–89% for periods below 50 h and 36% above it, where the method usually returns half of the true period. I report 34 asteroids that had no rotation period in LCDB, JPL SBDB, DAMIT, SsODNet, about 180 VizieR tables, or the Minor Planet Bulletin 2023–2026, as targets for confirmation. An independent reanalysis of all of Gaia DR3 with a refined method recovers the same period for all 34.

DATA AND METHOD

I ranked the numbered asteroids by the number of Gaia DR3 Solar-System-object observations (Tanga et al. 2023). From the top of this list I took the first 300 that have a rotation period of 2–100 h in JPL SBDB, which follows the October 2023 revision of the LCDB (Warner et al. 2009), as the validation set. I took the first 298 without a JPL period as the target set. Every selected asteroid has at least 443 Gaia observations. Gaia photometry was averaged per transit, with a 0.03 mag error floor. For every transit, JPL Horizons gave the heliocentric distance r, the distance Δ from Gaia and the phase angle α, with Gaia as the observer. Times were corrected for light time.

The reduced magnitude $G - 5\log_{10}(r\Delta)$ was fit by weighted least squares with one offset per observing season, a linear phase term and a second-order Fourier series in rotational phase. I scanned 2–100 h on a frequency grid with a step of $0.1/T$, where T is the time span. Because most asteroid lightcurves are double-peaked, I adopted the doubled period whenever doubling raised χ^2 by less than a factor of 1.5. This factor was chosen by two-fold cross-validation on the validation set. The confidence score combines the χ^2 gain over a no-rotation model with the separation of the best minimum from the next one outside the $\times 1$, $\times 2$ and $\times \frac{1}{2}$ aliases. The confidence threshold was set on the validation set, at the score where precision reaches 80%.

VALIDATION

The cross-validated recovery rate on the validation set is 67.7%. Above the threshold, I compared every result whose period is known (Figure 1):

- the validation set: 172 of 214 correct. This figure is in-sample, because the same set fixed the threshold, and is therefore optimistic;
- target-set asteroids that Ďurech & Hanuš (2023) solved from the same Gaia data with a different method: 81 of 97 agree. This tests the method, not the data;
- 25 of my initial candidates that later appeared in Cellino et al. (2024), MacLennan et al. (2026) or McNeill et al. (2023): 16 of 25 agree.

In total, 268 of 336 (80%) agree.

There are two failure modes. Above 50 h the method usually returns half of the true period (10 of 28 correct). Slow rotators with periods beyond 100 h, such as 214, 818 and about 4,600 h, receive spurious short periods.

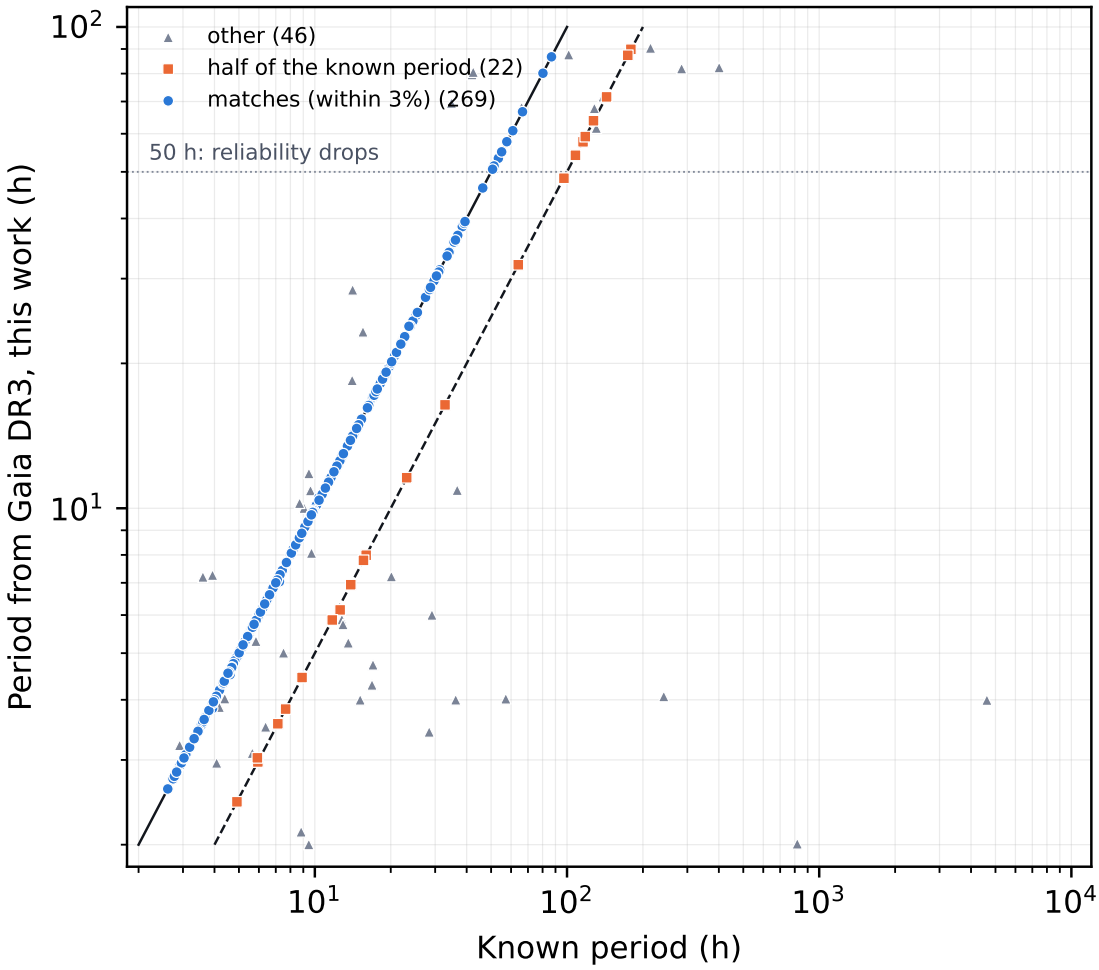


Figure 1. Period from Gaia DR3 (this work) against the published period for 337 confident results. The solid line marks agreement and the dashed line half of the known period. Above 50 h the method usually finds half of the true period.

RESULTS

The 34 candidates are all main-belt asteroids. In October 2026 they were absent from:

- JPL SBDB and LCDB V4.0;
- Ďurech & Hanuš (2023);
- DAMIT (Ďurech et al. 2010);
- the SsODNet compilation (Berthier et al. 2023), in a snapshot of its ssoBFT table;
- about 180 VizieR tables (5.1 million rows) that hold asteroid rotation periods;
- the full text of the Minor Planet Bulletin, volumes 50–53.

Of the 34 candidates, 30 have periods below 50 h, where I expect about 83% to be correct. Four have periods above 50 h.

An independent reanalysis of all numbered asteroids with at least 25 Gaia DR3 transits used a refined method (Tropynin 2026b). It recovers the same period, within 3%, for all 34. It also led me to drop seven earlier candidates:

- four with periods above 50 h, where the true period is most likely twice the value I found (7699, 14372, 43014, 104169);

- three where a slow solution is now preferred (13686, 39932, 109412).

The SsODNet check removed one more candidate: Ďurech et al. (2022) give 19.7181 h for 11857 from ATLAS photometry, which agrees with my 19.710 h.

Periods, alias periods, amplitudes, amounts of data and flags are available at https://github.com/olegtropinin/gaia-asteroid-periods and archived on Zenodo (Tropynin 2026a), together with the code and the validation tables. I offer the list as targets for dense-lightcurve confirmation.

This work used data from the ESA mission Gaia, processed by the Gaia DPAC; JPL Horizons and the JPL Small-Body Database; and VizieR (CDS, Strasbourg). The analysis was designed, coded and checked with the help of an AI assistant (Claude, Anthropic) under the author's direction.

REFERENCES

Berthier, J., Carry, B., Mahlke, M., & Normand, J. 2023, A&A, 671, A151

Cellino, A., et al. 2024, A&A, 687, A277

Ďurech, J., & Hanuš, J. 2023, A&A, 675, A24

Ďurech, J., Vávra, M., Vančo, R., & Erasmus, N. 2022, FrASS, 9, 809771

Ďurech, J., Sidorin, V., & Kaasalainen, M. 2010, A&A, 513, A46

MacLennan, E. M., et al. 2026, A&A, 707, A131

McNeill, A., et al. 2023, AJ, 166, 152

Gaia Collaboration, Tanga, P., et al. 2023, A&A, 674, A12

Tropynin, O. 2026a, Candidate rotation periods for main-belt asteroids from Gaia DR3 sparse photometry, Zenodo, doi: 10.5281/zenodo.23624115

Tropynin, O. 2026b, Rotation periods and spin senses of asteroids from Gaia DR3 sparse photometry, Zenodo, doi: 10.5281/zenodo.23638281

Warner, B. D., Harris, A. W., & Pravec, P. 2009, Icarus, 202, 134